

# Security and Privacy of Future Aircraft Wireless Communications with Offboard Systems

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**Abstract**—Modern global aviation must rise up to the challenge of safely, securely and efficiently managing increasingly crowded skies under growing passenger, business and societal demands. The “e-enabled aircraft” plays a central role in overcoming this grand challenge. Aeronautical-specific and commercial wireless data links as well as Internet standards will tightly integrate this aircraft with off-board systems in space, air and ground for revolutionizing air traffic control, flight operations and passenger services. However, potential vulnerabilities of these new technologies present risks to performance, user acceptance and deployment cost of future airspace systems. In this paper, we focus on these vulnerabilities and their mitigation. We discuss security and privacy threats and challenges that emerge with wireless-enabled air traffic control and flight operations.

**Keywords**—component; ATM, e-enabled aircraft, ADS-B, IPv6, security, privacy, anonymity

## I. INTRODUCTION

For almost a century, global air transport has evolved to an affordable, safe travel medium for society providing lucrative business for stakeholders. Most of today’s existing air transport infrastructure and processes, however, are decades old. They do not have the capacity to handle challenges of the next two decades, such as air traffic doubling, rapid fossil jet fuel depletion and noise and greenhouse gas emissions cap. The world’s largest Air Traffic Management (ATM) system in the USA, for instance, annually incurs over a million hours of delays, 38 million gallons of fuel burn, 60 million tonnes of CO<sub>2</sub> emissions, billions of dollars in unforeseen costs (<http://www.bts.gov>). Such alarming trends of aging and traffic congestion are emerging across the globe today. The persistent threat of terrorists constrains and challenges solutions.

Significant transformation of air transport systems is underway with the deployment of new infrastructure, technologies and processes, such as USA’s NextGen (<http://www.jpdo.gov/>) and Europe’s Single European Sky ATM Research (<http://www.eurocontrol.int/sesar>). Shift from radar-based to ADS-B surveillance is highly likely in India [1]. The future aviation system is envisioned as a large, complex, evolving ecosystem of airspace users, policies, procedures and technological enablers. Onboard systems of aircraft are anticipated to be very closely coupled with off-board systems for tighter control and monitoring of aircraft, airspace, airports and air traffic. The resulting capabilities promise unprecedented efficiencies for all phases of gate-to-gate flight.

A recent vision in commercial aviation is the “e-enabled aircraft” with advanced wireless technologies to operate as an intelligent mobile node in a global information network of systems in space, air and ground [2]–[4]. Global Navigation Satellite System (GNSS), Automatic Dependent Surveillance Broadcast (ADS-B), and wireless networking are major cornerstones of the future e-enabled aircraft [5], [6]. Together these offer higher automation, situational awareness and decision making for flight deck crew as well as air traffic controllers, to optimally perform ATM operations in time and space. Furthermore, advances in satellite and terrestrial data links promise seamless, global, broadband connectivity between aircraft and off-board systems on the Internet for airline operations and in-flight entertainment [2].

However, security and privacy risks with aircraft wireless communications will test future aviation systems performance gains. Recent reports highlight concerns with GNSS, ADS-B and airborne IP networking. Integrity, availability of GNSS must be highly assured for aircraft navigation, warranting various land-based augmentation solutions [7]. Federal Aviation Administration (FAA) mandates aircraft be ADS-B ready by year 2020 in the USA, but equipage, security and privacy remain as major risks [8]. Internet Protocol (IPv6) based airborne networking is emerging as large-scale challenge for global aviation [9]. Additionally, new challenges emerge when vulnerable commercial data links, e.g., IEEE 802.11 and 802.16, and cellular networks are expected to provide 24/7 availability and performance guarantees for airline and ATM operations at airports [10]. Existing solutions for wireless and mobile ad hoc networks do not directly apply to aviation problems due to unique properties such as high aircraft speeds, global operations, stringent regulatory requirements, and airspace constraints (see Section II.A).

## A. Paper Contributions and Outline

We present a framework to address security and privacy of future wireless communications between aircraft and off-board systems. Due to the safety and business criticality of flight deck communications, we specifically focus on major vulnerabilities of ADS-B that potentially threaten airspace security and airspace user privacy. We present threat mitigation solutions that make use of physical properties of airspace and air traffic.

Section II presents the security and privacy framework. Sections III–VI apply this framework to ADS-B and discuss several open problems. Section VII concludes the paper.

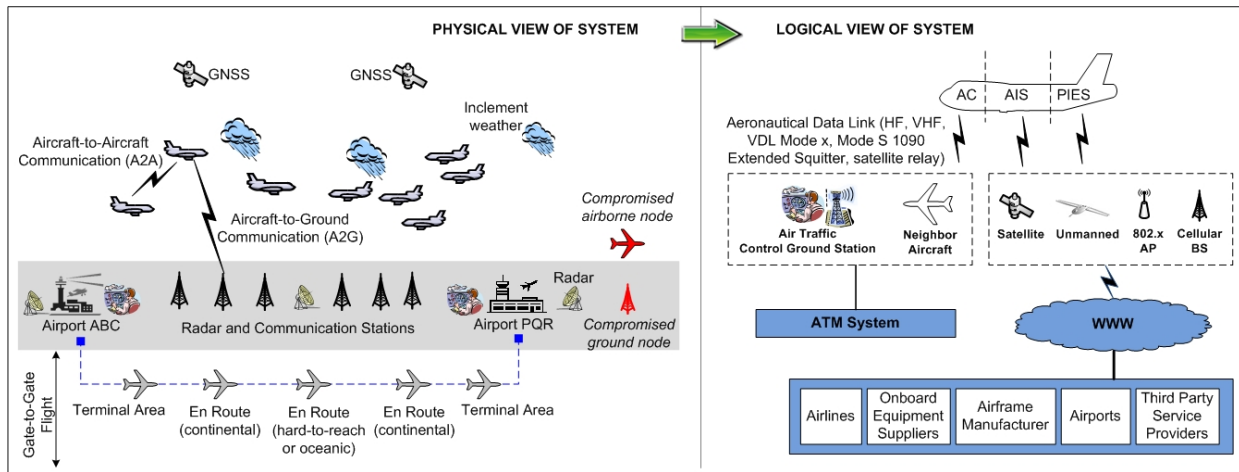


Figure 1. Illustration of the future airspace system model with aircraft wireless communications and stakeholders.

## II. PROPOSED SECURITY & PRIVACY FRAMEWORK

### A. Airspace System Properties and Constraints

Fig. 1 shows an envisioned civilian airspace system that is extremely complex with heterogeneous types of e-enabled aircraft, distributed infrastructures and systems-of-systems on the ground [2]. The airspace is classified into different classes based on factors such as location, use and flight rule restrictions, e.g., Class A to Class G in the United States [11]. Airspace usage restrictions apply based on aircraft type and extraneous constraints such as weather and air traffic congestion. System performance, such as safety, capacity and efficiency, heavily rely on unforgiving environment conditions, e.g., inclement weather, poor visibility, icing, turbulence and winds, often difficult to model and predict. Furthermore, for airspace security all aircraft must be identifiable, accessible, controllable and uniquely traceable in the system.

Fig. 1 shows a commercial aircraft's flight classified into departure/arrival, terminal, and en-route phases. This gate-to-gate flight includes aircraft parked, taxiing, takeoff/landing, or cruising in radar and non-radar covered airspaces. Aircraft movement is spatially and temporally constrained by tactical (e.g., inter-aircraft separation) and strategic air traffic rules (e.g., pre-planned routes and traffic flows). Aircraft routes hence are usually predictable, except in some en-route, uncontrolled airspace, weather and traffic congestion impacted regions, with a pre-flight established trip time. Hence aircraft tend to travel as a geo-proximate group in airspace.

Civil aviation is subject to well-established stringent regulations to assure that aircraft safety and public well being are protected in the airspace system. Regulatory agencies, such as the Federal Aviation Administration (FAA), set mandatory guidelines for airlines to ensure airworthiness of operational aircraft as well as minimum performance standards and rules for ATM tasks. Regulations are emerging for aircraft environmental performance, i.e., airport noise/air quality, GHG emissions (<http://www.boeing.com/commercial/noise>), [12]. Under such constraints, any new ground or airborne technology must be shown to not adversely impact safe, expected operations, under nominal and off-nominal conditions.

Interests of stakeholders involved in the aircraft life cycle from design to disposal, may often compete due to conflicting business goals and limited airspace system resources. For instance, while a fuel-efficient route may benefit airlines, air traffic controllers and passengers may prefer a delay-optimal route. Airspace system equipage, operation and maintenance costs, return-of-investment on existing technology and processes are also critical business factors.

### B. Classification of Aircraft Off-board Communications

Fig. 1 shows aircraft-to-ground communications (A2G) and aircraft-to-aircraft communications (A2A) are used to closely monitor and control aircraft status, movement and operations in the airspace system. Regulatory reasons and continuity/longevity of onboard avionics, however, warrant the various A2G and A2A communications are treated differently as well as use distinct data links and service providers [13]. Fig. 1 shows the resulting classification of off-board communications as: Aircraft Control (AC), Airline Information Services (AIS), and Passenger Information and Entertainment Services (PIES).

AC communications include voice and data exchanges with air traffic controllers, neighbor aircraft, airlines and other stakeholders for flight safety, regularity and efficiency. This includes ATM, in-flight security (i.e., cockpit, cabin sensor data) and airline operational communications performed via aeronautical protocols over different RF links, i.e., satellite relay, HF, VHF Data Link (VDL) Mode 2/3/4, Mode-S 1090 MHz Extended Squitter (1090-ES), dedicated IEEE 802.16e, L-band terrestrial links [2], [4], [12]. Each aircraft will include a globally unique digital identity in these communications, such as the 24-bit International Civil Aviation Organization (ICAO) address, tail or flight number [14]. The enabled ATM exchanges help aircraft fly in 4-dimensional routes (i.e., 3-D position and time) and allow monitoring of positions, movement and conformance of aircraft on these routes. Observed airspace dynamics, tactical or strategic changes in flight route and schedule are all timely coordinated with air traffic controllers and airlines. Furthermore, traffic and flight information can be sent to aircraft, e.g., weather update, locations of legacy aircraft, or temporary airspace restrictions.

AIS communications includes business-critical voice and data exchanges for flight logistics, aircraft maintenance, etc. For these communications, aircraft is connected to the Internet, over satellite or unmanned system relays during en-route, emerging IEEE 802 wireless (e.g., 802.11, 802.16) or cellular networks at airports. Aircraft interacts with airlines, service providers and other stakeholders to seamlessly refresh and distribute onboard digital content, such software updates and sensor data. PIES communications enable services such as in-flight entertainment (e.g., on-demand video, live television), passenger health and passenger Internet access. In the future, broadband IP connectivity at high altitudes and remote regions will become a key need for AIS and PIES [2]. Multi-hop IP networking over A2A links further promises enhanced coverage in remote regions for off-board communications [12].

Regulatory restrictions limit the use of off-board communications to certain flight phases. AC communications is used throughout flight. During cruise, broadband communications over satellite, unmanned aircraft or VHF data links become available for all off-board communications. At airport gates, large volume data transfers over commercial data links are allowed. Furthermore, performance constraints and criticality of aircraft wireless communications vary based on air traffic and flight conditions. For instance, high airspace density or convective weather present radio interference conditions as well as integrity and availability requirements for aircraft wireless communications at and around airports.

### C. Adversary Model Considered

The adversary aims to degrade performance of the airspace system by creating unforeseen disruptions and behaviors. The adversary is capable of accessing and misusing air traffic information assets, e.g., motivation of hackers or criminal organizations. The adversary can be external to the system (e.g., an unauthorized transponder on the ground), capable of passive eavesdropping of communications as well as active attacks on communications, i.e., data corruption (e.g., create intentional errors during message transmission, transmit bogus messages), spoofing (e.g., forge identity in message, replay authentic messages) and wireless channel jamming (i.e., degrade signal-to-noise ratio at the receiver). Furthermore, the adversary can be an insider who compromises, for example, some unattended ground stations or aircraft for active attacks.

Insider access to aircraft can be assumed deterred by legal regulations and sufficiently safeguarded against using physical, logical and organizational inhibitors. However, this paper considers threats from external attackers as well as insiders, for rigor and completeness of security analysis. Such an approach can enhance the level of protection to future airspace systems.

### D. Threats to Aircraft Wireless Communications

1) *Corruption of information assets*: The adversary may attempt to corrupt certain critical data to create unwarranted concerns about aircraft safety or create false alarms and late detection of corruption that delay flights and cost airlines/passengers. Some examples include AC data communications, such as ADS-B message, avionics software updates or flight bag used in flight operation and management, health diagnostics used to detect/monitor faults [2].

2) *Misuse of information assets*: Communicated assets may provide knowledge useful for the adversary in side channel attacks, e.g., aircraft locations or fuel levels. Certain assets may also be considered to be intellectual property with business value, e.g., RFID tag or engine sensor data which may contain some sensitive part maintenance data [2].

3) *Delay of information assets*: Time-critical assets may become inaccessible due to intentional jamming of aircraft wireless communication channels, potentially degrading the performance of the airspace system.

4) *Repudiation*: An entity in the airspace system could deny having sent or received information assets after detection of corruption or misuse, thereby disrupting liability for accident or failure events in the airspace system.

### E. Security Primitives for Aircraft Wireless Communications

1) *Integrity and Authenticity*: For preventing any corruption of assets by the adversary, the identity and content of the asset received at the destination must be verified to be the same as at the source. Further, the source identity must also be verified.

2) *Confidentiality, Communication Anonymity and Location Untraceability*: Unauthorized access to sensitive assets that can be leveraged for future attacks or personal gain must be prevented. Broadcasts from an aircraft must not be linkable to the aircraft's digital identifier by unauthorized entities. Furthermore, two consecutive broadcasts from an aircraft cannot be linked together by unauthorized entities.

3) *Availability*: Each communicated asset must be available soon enough to meet aircraft time constraints. Further, minimum performance standards of the airspace system must be met in the event of unavailable communications.

4) *Airspace Security and Liability*: The integrity and timely availability of aircraft communications must be guaranteed. Furthermore, communications from an aircraft must be irrevocably linkable to that aircraft by authorized entities.

The remainder of the paper focuses on using this framework to evaluate a specific AC communications, called ADS-B, for monitoring of aircraft movement in ATM systems.

## III. ADS-B BASED MONITORING OF AIRCRAFT

ADS-B brings a paradigm shift from surveillance radar based monitoring and control of aircraft movement in ATM systems. Fig. 2 illustrates the basic idea of ADS-B. An aircraft transponder (i) computes its 3-dimensional position using GNSS and other spatial information, e.g., speed, heading intent, from onboard navigational sensors (ii) constructs a message that contains the derived position and time, other spatial information and unique aircraft identifier, (iii) uses a data link to broadcast the constructed message, and (iv) repeats above steps to broadcast a message with updated information periodically. The resulting frequent, widely available, accurate position feedback from aircraft, among other benefits, reduces inter-aircraft separation minima in continental as well as oceanic airspaces. Such a spacing reduction translates to airspace system performance gains in terms of capacity and efficiency, both in radar and non-radar airspaces.

Existing ADS-B data links include 978 MHz Universal Access Transceiver (UAT) and Mode-S 1090 MHz Extended Squitter (1090-ES). Achievable communication range with these links is well over 100 miles [12], [14]. Message period varies between 0.5 to 1 second [14]. Default aircraft identifier in the message is assumed to be a permanent digital identifier such as the 24-bit International Civil Aviation Organization (ICAO) identity uniquely assigned to each operational civil aircraft, which helps maintain airspace security and liability.

Fig. 2 shows that ADS-B has two modes. One is called *ADS-B-Out* which makes an aircraft periodically broadcast position messages and traffic controllers on the ground can use these messages for *ground surveillance*, i.e., monitor airspace with high accuracy and integrity of GPS as well as wide coverage of the data link. Another is the a revolutionary mode called *ADS-B-In*. It offers *airborne surveillance* capability for an aircraft to receive and use position messages from neighboring nodes in airspace and airports, improving pilot situational awareness and alertness. In the longer-term, it helps self separation assurance and spacing between aircraft. Future ATM heavily relies on the accuracy and performance of ADS-B surveillance, for example, fuel and time efficient operations at/near high density airports, en-route and remote airspace.

However, potential security vulnerabilities in both GNSS and ADS-B technology remain to be assessed and resolved, impeding rapid design, deployment and beneficial usage of many ADS-B applications [8], [15], [16].

#### IV. ADS-B VULNERABILITIES AND THREATS

Vulnerabilities in ADS-B ground and airborne surveillance applications emerge because of the dependency of critical ATM tasks on data “openly” delivered on shared communication link. We identify following potential ADS-B vulnerabilities that result in threats classified in Section II.D.

##### A. Threat of ADS-B Message Corruption

The adversary may attempt to remotely manipulate GNSS readings to introduce errors in position computations onboard. Further, ADS-B position data may be manipulated or spoofed over the data link, generating false aircraft position claims.

##### B. Threat of ADS-B Message Misuse

ADS-B messages can be misused by the adversary to identify, locate and track a target aircraft. Such unauthorized location tracking raises potential privacy concerns for some airspace users, e.g., business jets, chartered and personally owned aircraft. Location trajectories of a private aircraft, when correlated with public databases and geographic maps, can help in accurately inferring business or political developments, travel intent and profiling of personal location preferences and interests. Such tracking may also expose high-value aircraft, e.g., law enforcement, to physical threats from ground.

##### C. Threat of ADS-B Message Delay

The adversary may attempt to intentionally jam wireless communications in a given region-of-interest, resulting in non-availability of ADS-B services and/or GNSS outages.

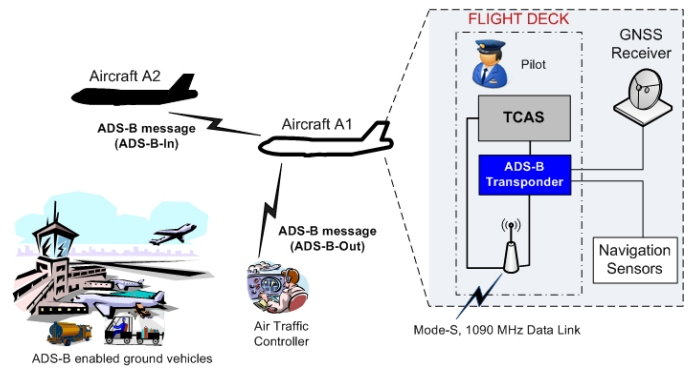


Figure 2. Illustration of ADS-B and its operational modes.

#### V. MITIGATION OF THREATS TO ADS-B

##### A. Mitigating Threats to GNSS Integrity

Accuracy of GPS readings can be achieved by making each aircraft include spatial precision parameters in ADS-B-Out message to establish error margins for computations at the receivers [17]. Unintentional GPS degradations around airports can be compensated using the Ground Based Augmentation System or GBAS which operates in the VHF range and transmits differential corrections in GPS receiver measurements to aircraft approaching or leaving the airport [7].

##### B. Mitigating Threats to Integrity of ADS-B-Out Messages

Primary surveillance radars on the ground and Traffic Collision Avoidance System (TCAS) onboard aircraft can serve as robust mechanisms to interrogate a compromised aircraft and verify its ADS-B position. Such an approach, however, requires ability to fuse data from radar and ADS-B as well as integration of TCAS and ADS-B. Furthermore, multilateration serves as a well-established solution for verifying position of an aircraft. Three or more verifiers can combine estimates of their relative distance to the aircraft, and bound the 2-D position of the aircraft to be at the intersection of three circles. 3-D position verification requires four or more verifiers on the ground. Several multilateration based mechanisms have been proposed to verify position information in ADS-B data received at the ground controllers [19], [21]. Multilateration in some situations however may have weaknesses. For example, it may be possible to simulate virtual ADS-B transmitters by varying timing of multiple transmitters at a given position or offset the position calculation by disrupting clock synchronization of the ground receivers [15]. Furthermore, symmetric and asymmetric key based cryptographic approaches can help protect ADS-B-Out messages [20]. A major challenge is however scalability of the cryptographic key infrastructure employed.

##### C. Mitigating Threats to ADS-B Message Anonymity

ADS-B is inherently open broadcast of position messages, hence making message confidentiality irrelevant for most airspace users. However, for anonymity, each ADS-B message must not be linkable to the permanent identity of private aircraft by unauthorized entities. A solution approach is to generate a 24-bit pseudo-random identifier (called pseudonym)



for the aircraft. A practical solution is outlined as a standard in [14]. The private aircraft computes the pseudonym as a function of a random quantity, e.g., the location and/or time of use of pseudonym, and the 24-bit ICAO identifier. Because air traffic controllers know the ICAO address of the aircraft as well as can record and verify ADS-B broadcasts from the aircraft, they can maintain airspace security and liability.

The use of pseudonyms however cannot provide location untraceability since there is strong spatial and temporal correlation between aircraft locations due to underlying predictable mobility of aircraft and the short inter-message period of ADS-B (e.g., 0.5 seconds).

#### D. Mitigating Threats to ADS-B Availability

A high-level assessment of potential impact of GNSS vulnerabilities, such as intentional jamming interference, on aviation safety is in [7]. Detectable failure or outages of GNSS can be addressed by employing secure land-based alternatives as a backup, such as terrestrial radio navigation systems operating in low-frequency range (e.g., e-LORAN) [21]. In radar-covered areas, primary/secondary surveillance radars can serve as land-based alternatives for sourcing aircraft position information during potential failures of GNSS and ADS-B [8]. Upon detection of an outage or disruption, a strategy is needed to gracefully degrade ADS-B based surveillance to a less accurate coverage of the backup system. As discussed in [18], the “circle of avoidance” of an aircraft must be gradually increased over a short time period to ensure safe transition to the degraded surveillance accuracy. Recovery strategy for outages in non-radar covered regions is also an open problem.

A major research gap however is the current lack of mitigation support for ADS-B-In message integrity and ADS-B message anonymity and location untraceability, addressed next.

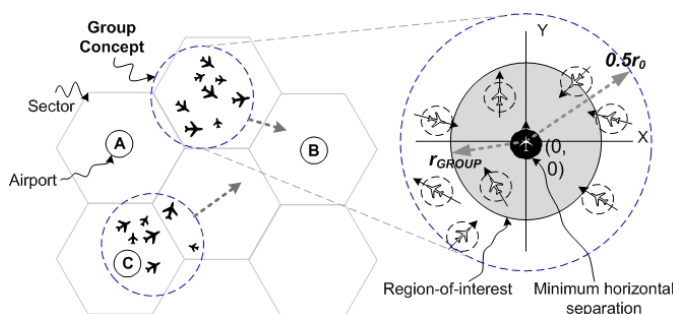


Figure 3. 2-D view of proposed group concept for ADS-B security

## VI. PROPOSED MITIGATION SOLUTIONS FOR ADS-B

### A. Building Block of the Proposed Solutions

Aircraft mobility is spatially/temporally restricted and spatially dependent resulting in the group navigation property, i.e., aircraft in geographical proximity navigate as a group for a

time period with same average velocity and similar direction. Furthermore, a group of navigating aircraft within distance  $0.5r_0$ , where  $r_0$  is aircraft IP network link range: (i) are geographically proximate, (ii) can maintain a mobile, fully connected IP network, and (iii) do not suffer any significant signal degradation in group communications [15]. Based on the above observations, the *group concept* is proposed where neighboring aircraft form a reliable communications group during navigation. Each Group Member must be able to hear broadcast of every other member. Such a group can be represented by a single aircraft, i.e., Group Leader. The full IP connectivity allows group communications for members to establish shared secrets, if needed, and perform joint operations such as validating ADS-B-In messages.

Fig. 3 shows an aircraft located at the center safely navigating in airspace shared with a group of neighboring aircraft. Since group operations incur overhead for each group member and due to factors such as communication interference and pilot workload, all neighbors within  $0.5r_0$  (e.g., aircraft within 50 miles) need not be members of a single group. The region of interest can be limited to  $r_{GROUP}$ . The required minimum horizontal separation between aircraft can vary from 3 to 50 nmi based on the flight phase and airspace region of the aircraft. Fig. 4(a) shows average group size expected for typical airspace densities and communication ranges in the USA.

### B. Leverage Group Concept for ADS-B-In Integrity

When an ADS-B signal from an aircraft P is received by four or more aircraft  $\{V\}$  in a group G, then  $\{V\}$  can use time-difference-of-arrival technique to estimate the 3-D position of P. The aircraft P can be outside G, or belong to G but acting erroneously for a time period T. In this algorithmic technique,  $\{V\}$  must measure time-of-arrival of the ADS-B signal from P, securely share this time-of-arrival with each other, compute the differences between times, then perform multilateration computations to estimate actual position of P. An underlying principle of this multilateration computation is that RF signals cannot be sent faster than the speed of light  $c$  [22]. Therefore,  $\{V\}$  can independently estimate P's actual position and verify if claimed position of the ADS-B-In message from P matches the actual position of P. Furthermore, if high assurance is needed for this verification,  $\{V\}$  can perform secure voting.

Instead of time-of-arrival, each aircraft in  $\{V\}$  can estimate relative distance to  $P$  using received signal strength. However, if the adversary is capable of varying its transmission power, the verifier does not have knowledge of the transmitted signal strength and cannot estimate the relative distance based on the received signal strength. Moreover, distance measurements are susceptible to distance enlargement or reduction [22].

Figure 4(b) shows the probability of successful position estimation and verification by proposed solution for different traffic densities and aircraft communication range of 60 nmi.

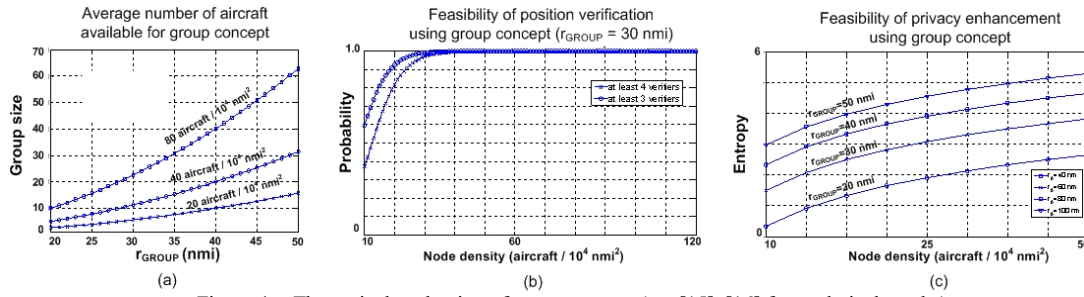


Figure 4. Theoretical evaluation of group concept (see [15], [16] for analytical results).

### C. Leverage Group Concept for ADS-B Privacy

The group of aircraft can continue to broadcast ADS-B-Out messages with pseudonyms, while coordinating to be represented by a common valid group identifier for IP communications as well as establishing secrets on-the-fly for group-based operations. Each aircraft in a group reduces its transmission range to reach only other members of the group (such as 6 – 10 nmi). Each group has a leader with a large transmission range that is sufficient to reach airborne and ground transponders (such as 100 nmi). A candidate for group leader is an aircraft that does not require location privacy.

In such a solution, adversary can at best know group's identifier and leader's location. Group identifier is only traceable to a navigating set of aircraft and members can self update their pseudonyms while participating in the group. Each member potentially achieves an extended random time period for the pseudonym update, equivalent to duration for which the member remains in the group. This potentially decorrelates two consecutive locations of a target observed by the adversary [16]. The ground controllers can still identify, accurately trace valid nodes, while unauthorized eavesdroppers outside the group can only randomly guess aircraft trajectories. Fig. 4(c) shows average location privacy level (entropy measure [16]) for various airspace densities, aircraft communication ranges.

## VII. CONCLUSIONS AND FUTURE WORK

We presented a security and privacy framework for future aircraft wireless communications in airspace systems. We focused on applying this framework to ADS-B. We classified security and privacy threats that emerge from ADS-B vulnerabilities and presented mitigation solutions. Two challenging open problems include providing message integrity for ADS-B-In and mitigating unauthorized location tracking of ADS-B equipped private aircraft. We proposed preliminary solution approaches to these two problems by leveraging group navigation of aircraft in the airspace system. Future work will develop these solutions as well as graceful degradation of ATM system performance in the presence of intentional jamming.

## ACKNOWLEDGEMENTS

Any findings, conclusions or recommendations in this material are those of the authors, and should not be interpreted as of The Boeing Company.

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